ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

Unit 6.01: Permutations & Combinations

Q1. Evaluate the number of permutations {}^{5}P_{2} .

[ ] A: 25 [ ] B: 21
[ ] C: 10 **[x] D: 20**

Solution: {}^{n}P_{r} = frac{n!}{n-r!} implies {}^{5}P_{2} = 5 times 4 = 20 .

Q2. Evaluate the number of permutations {}^{6}P_{2} .

[ ] A: 12 [ ] B: 15
[ ] C: 36 **[x] D: 30**

Solution: {}^{n}P_{r} = frac{n!}{n-r!} implies {}^{6}P_{2} = 6 times 5 = 30 .

Q3. Evaluate the number of permutations {}^{7}P_{2} .

[ ] A: 21 **[x] B: 42**
[ ] C: 49 [ ] D: 14

Solution: {}^{n}P_{r} = frac{n!}{n-r!} implies {}^{7}P_{2} = 7 times 6 = 42 .

Q4. Evaluate the number of permutations {}^{8}P_{2} .

[ ] A: 64 **[x] B: 56**
[ ] C: 16 [ ] D: 28

Solution: {}^{n}P_{r} = frac{n!}{n-r!} implies {}^{8}P_{2} = 8 times 7 = 56 .

Q5. Evaluate the number of permutations {}^{9}P_{2} .

[x] A: 72 [ ] B: 81
[ ] C: 18 [ ] D: 36

Solution: {}^{n}P_{r} = frac{n!}{n-r!} implies {}^{9}P_{2} = 9 times 8 = 72 .

Q6. Evaluate the number of permutations {}^{10}P_{2} .

[x] A: 90 [ ] B: 100
[ ] C: 20 [ ] D: 45

Solution: {}^{n}P_{r} = frac{n!}{n-r!} implies {}^{10}P_{2} = 10 times 9 = 90 .

Q7. A committee of 4 people is to be chosen from 5 men and 4 women. How many committees consist of exactly 2 men and 2 women?

[x] A: 60 [ ] B: 120
[ ] C: 55 [ ] D: 36

Solution: binom{5}{2} times binom{4}{2} = frac{5times4}{2} times frac{4times3}{2} = 60 .

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Q8. A committee of 4 people is to be chosen from 6 men and 5 women. How many committees consist of exactly 2 men and 2 women?

[ ] A: 44

[ ] B: 145

[ ] C: 300

[x] D: 150

Solution: $\binom{6}{2} \times \binom{5}{2} = \frac{6\times5}{2} \times \frac{5\times4}{2} = 150$.